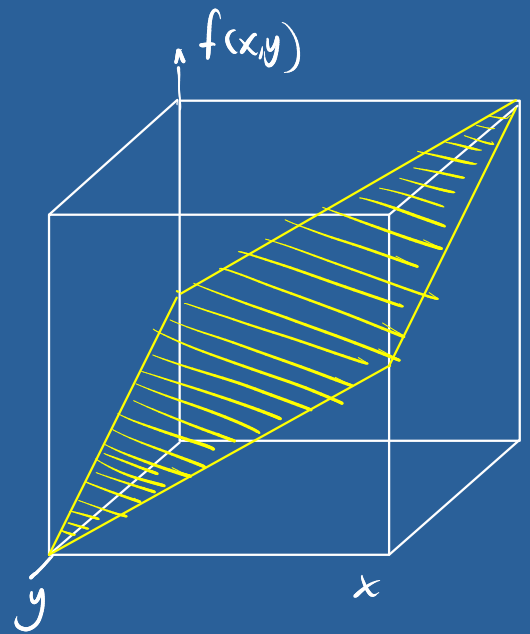


SOLUCIÓN PROBLEMA 2.

$$f_{xy}(x,y) = \begin{cases} \frac{2}{3}(x+2y), & 0 \leq x \leq 1 \\ 0, & \text{en cualquier otro caso} \end{cases}$$



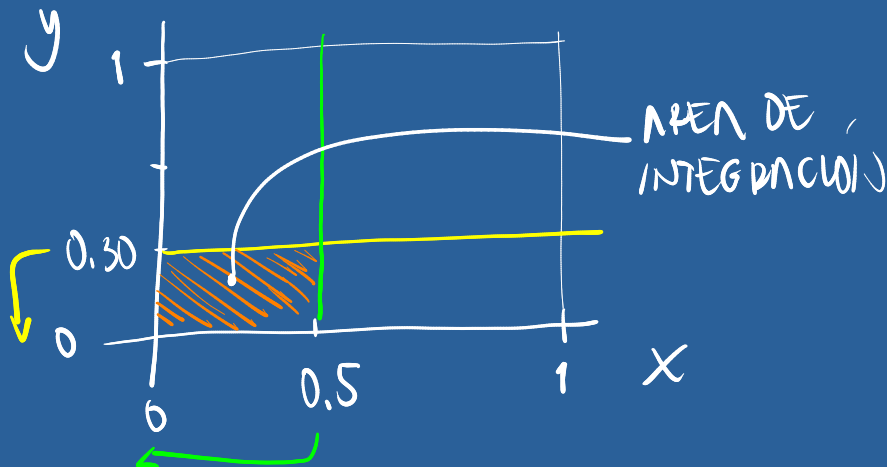
$$a) \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x,y) dx dy = 1$$

$$= \int_0^1 \int_0^1 \frac{2}{3}(x+2y) dx dy = \int_0^1 \left(\frac{2}{3} \left(\frac{x^2}{2} + 2xy \right) \Big|_0^1 \right) dy$$

$$= \int_0^1 \frac{2}{3} \left(\frac{1}{2} + 2y \right) dy = \frac{2}{3} \left(\frac{y}{2} + \frac{2y^2}{2} \right) \Big|_0^1 = \frac{2}{3} \left(\frac{1}{2} + 1 \right) = 1 //$$

SE CUMPLE
CONDICIÓN — OK

$$b) P(X \leq 0.5 ; Y \leq 0.30) = \int_0^{0.30} \int_0^{0.50} \frac{2}{3}(x+2y) dx dy$$



$$= \int_0^{0.30} \left(\frac{2}{3} \left(\frac{0.5^2}{2} + y \right) \right) dy = \int_0^{0.30} \frac{2}{3} (0.125 + y) dy$$

$$= \frac{2}{3} \left(0.125 y + \frac{y^2}{2} \right) \Big|_0^{0.30} = 0.055 //$$

$P(X \leq 0.80) \rightarrow$ se debe hallar $g(x)$ FUNCIÓN DE DENSIDAD MARGINAL DE X

$$g(x) = \int_{-\infty}^{\infty} f(x,y) dy = \int_0^1 \frac{2}{3}(x+2y) dy$$

$$= \frac{2}{3} \left(xy + \frac{2y^2}{2} \right) \Big|_0^1$$

$$g(x) = \begin{cases} \frac{2}{3}(x+1), & 0 \leq x \leq 1 \\ 0 & \text{en cualquier otro caso} \end{cases}$$

$$= \frac{2}{3}(x+1)$$

$$P(X \leq 0.80) = \int_0^{0.80} g(x) dx = \int_0^{0.80} \frac{2}{3}(x+1) dx$$

$$= \frac{2}{3} \left(\frac{x^2}{2} + x \right) \Big|_0^{0.80} = 0.7467 //$$

$P(Y \geq 0.60)$ — debemos hallar $h(y)$ FUNCIÓN DE DENSIDAD MARGINAL DE Y

$$h(y) = \int_{-\infty}^{\infty} f(x,y) dx = \int_0^1 \frac{2}{3}(x+2y) dx$$

$$= \frac{2}{3} \left(\frac{x^2}{2} + 2xy \right) \Big|_0^1$$

$$h(y) = \begin{cases} \frac{2}{3} \left(2y + \frac{1}{2} \right), & 0 \leq y \leq 1 \\ 0, & \text{en cualquier otro caso} \end{cases} = \frac{2}{3} \left(\frac{1}{2} + 2y \right)$$

$$P(Y \geq 0.60) = 1 - P(Y < 0.60) = 1 - \int_0^{0.60} \frac{2}{3} \left(2y + \frac{1}{2} \right) dy$$

$$= 1 - \left(\frac{2}{3} \left(\frac{2y^2}{2} + \frac{y}{2} \right) \right) \Big|_0^{0.60} = 1 - \left(\frac{2}{3} \left(0.60^2 + \frac{0.60}{2} \right) \right)$$

$$= 0.56$$

$$c) \rho_{XY} = \frac{\text{COV}(XY)}{\sqrt{V(X) V(Y)}}$$

$$\text{COV}(XY) = E(XY) - E(X)E(Y)$$

$$E(Y^2) - E(Y)^2 \quad \int_{-5}^5 \int_{-5}^5 xy f(xy) dx dy$$

$$E(X^2) - E(X)^2 \quad \int_{-5}^5 y h(y) dy$$

$$\int_{-5}^5 x g(x) dx$$

$$E(X) = \int_0^1 x \left(\frac{2}{3}(x+1) \right) dx = \int_0^1 \frac{2}{3}(x^2 + x) dx = \frac{2}{3} \left(\frac{x^3}{3} + \frac{x^2}{2} \right) \Big|_0^1$$

$$= \frac{2}{3} \left(\frac{1}{3} + \frac{1}{2} \right) = \frac{2}{3} \left(\frac{5}{6} \right) = \frac{5}{9} //$$

$$E(X^2) = \int_0^1 x^2 \left(\frac{2}{3}(x+1) \right) dx = \int_0^1 \frac{2}{3}(x^3 + x^2) dx =$$

$$\frac{2}{3} \left(\frac{x^4}{4} + \frac{x^3}{3} \right) \Big|_0^1 =$$

$$\frac{2}{3} \left(\frac{1}{4} + \frac{1}{3} \right) = \frac{2}{3} \left(\frac{7}{12} \right) = \frac{7}{18} //$$

$$V(X) = \frac{7}{18} - \left(\frac{5}{9} \right)^2 = \frac{13}{162} //$$

$$\begin{aligned}
 E(Y) &= \int_0^1 y h(y) dy = \int_0^1 y \left(\frac{2}{3} \left(2y + \frac{1}{2} \right) \right) dy = \int_0^1 \frac{2}{3} \left(2y^2 + \frac{y}{2} \right) dy \\
 &= \frac{2}{3} \left(\frac{2y^3}{3} + \frac{y^2}{4} \right) \Big|_0^1 = \frac{2}{3} \left(\frac{2}{3} + \frac{1}{4} \right) \\
 &= \frac{2}{3} \left(\frac{11}{12} \right) = \frac{11}{18},
 \end{aligned}$$

$$\begin{aligned}
 E(Y^2) &= \int_0^1 y^2 h(y) dy = \int_0^1 y^2 \left(\frac{2}{3} \left(2y + \frac{1}{2} \right) \right) dy = \\
 &\int_0^1 \frac{2}{3} \left(2y^3 + \frac{y^2}{2} \right) dy = \frac{2}{3} \left(\frac{2y^4}{4} + \frac{y^3}{6} \right) \Big|_0^1 \\
 &= \frac{2}{3} \left(\frac{1}{2} + \frac{1}{6} \right) = \frac{2}{3} \left(\frac{4}{6} \right) = \frac{4}{9},
 \end{aligned}$$

$$V(Y) = \frac{4}{9} - \left(\frac{11}{18} \right)^2 = \frac{23}{324}$$

$$\begin{aligned}
 E(XY) &= \int_0^1 \int_0^1 xy f(xy) dx dy = \int_0^1 \int_0^1 \frac{2}{3} (x^2y + 2xy^2) dx dy \\
 &= \int_0^1 \frac{2}{3} \left(\frac{yx^3}{3} + \frac{2y^2x^2}{2} \right) \Big|_0^1 dy = \int_0^1 \frac{2}{3} \left(\frac{y}{3} + y^2 \right) dy \\
 &= \frac{2}{3} \left(\frac{y^2}{6} + \frac{y^3}{3} \right) \Big|_0^1 = \frac{2}{3} \left(\frac{1}{6} + \frac{1}{3} \right) = \frac{2}{3} \left(\frac{3}{6} \right) = \frac{1}{3}
 \end{aligned}$$

$$\begin{aligned}
 \text{COV}(XY) &= E(XY) - E(X)E(Y) \\
 &= \frac{1}{3} - \frac{5}{9} \times \frac{11}{18} = -\frac{1}{162} \approx -0.00617 //
 \end{aligned}$$

$$\rho_{xy} = \frac{\text{COV}(XY)}{\sqrt{V(X)V(Y)}} = \frac{-0.00617}{\sqrt{\frac{13}{162} \times \frac{23}{324}}} \approx -0.0817 //$$

RELACION
NEGATIVA
MUY DEBIL //